

What needs to change, why, and how

Checkpoint-2 memo for *Escalate or Answer* (design v3)

For Coen Petto and Cameron Suess · CS 567, Fall 2026 · prepared 28 September 2026

Bottom line

The instructor’s verdict is right and the fix is smaller than it looks. Keep the interest (an agent that hands a decision back to a person), keep the within-subjects machinery, the replayed traces, the FictionalQA filter and the MiPP anchor, and drop what could not be run: the expert population, the learning post-test, the invented corpus, and the “escalates vs. never escalates” manipulation. What replaces them is Option 1: fix the escalation points, vary the *form* of the escalation (modal stop, inline flag, provenance-only) within subjects, on a frozen snapshot of the CSU CS systems documentation, with students new to that system. The escalation points are *context-dependent forks*: two pages give different answers, both correct for someone, and which applies depends on a fact about the participant that the assistant was never told. On each trial the assistant streams a short cited procedure and pre-fills a card of dropdowns; the participant may check the pages, then submits. Whether they checked, whether they kept or changed the agent’s default when it was right and wrong, and what it cost are what is measured. The redesigned proposal is **new_proposal.pdf** (about five pages including the mock-ups); this memo explains every change and what you still have to do by hand.

1 Framing and title

What	Why	How
Retitle as a question, and fit the new fork type.	The old title asserted the result; the instructor’s C1 asks for the question. “When its sources conflict” no longer fits: both pages are right, for different users.	“Escalate or Answer: How Should an Agentic RAG Assistant Hand Back a Decision That Depends on the User?” Keep “agentic” only while the four conditions in the answer box hold.
One study, one RQ.	Two experiments on two populations with two DVs doubled the work and neither was runnable.	RQ: when the assistant finds mid-task that the right answer depends on an unstated fact about the user and hands the choice back, how does the form of that hand-back affect verification, appropriate reliance and cost?
Author block without role labels; roles in one “Team” paragraph.	The rubric asks for the group name, members and what each person does; the instructor said role labels must not sit next to author names and that roles leave the PDF from checkpoint 3.	Two \author lines with \email, one affiliation. A three-line Team paragraph at the end of Deliverables (Coen: study design, web app, sessions; Cameron: pipeline, item authoring, corpus snapshot), marked <code>TODO authors: confirm roles</code> . Delete it at checkpoint 3.
Fix “these three papers”.	No antecedent.	Gone; every “this” and “these” now has a noun after it.

2 Why the “newer page wins” rule was dropped (new in v3)

What	Why	How
Replace date conflicts with context-dependent forks.	Version 2 keyed every conflict on “last updated” dates. Coen identified the flaw and it is real: (1) a competent agent could resolve a date conflict itself, so escalating it is bad agent design and a reviewer would say “fix the agent” (MiPP and Mishra justify handing a decision to a human only when the system cannot resolve it); (2) it turned verification into a mechanical instruction-following task (compare two dates); (3) it did not fit the RQ, which is about when a human is actually needed.	Escalate only forks the documents cannot settle alone: two pages give different answers, each correct for someone, and which applies depends on a fact about the user the question does not state. Example: GPU Jobs lists RTX 3090s (24 GB) on kestrel-gpu and A100s (40/80 GB) on peregrine-gpu ; “set up a GPU job for my PyTorch model” does not say how much memory the model needs.
Split the task card: “Your situation” / “What you asked”.	The deciding fact must exist for the participant but not for the assistant, and people ask short questions.	<i>Your situation</i> : one or two lines only the participant sees (“your model needs about 30 GB”). <i>What you asked the assistant</i> : the short question, the only input the assistant receives.
Keys from page scope plus situation; no dates, no authority rule, no rule card.	The key must be derivable from the frozen snapshot alone (no time or money for administrator verification) and must not depend on a stated rule.	Each page states its own scope (memory per GPU; “off campus, use the VPN”; “not backed up, scrubbed”). Key = page scope × situation line. Default = the higher-ranked retrieval option. Default-correct items reuse the same pages with a different situation line (a model that fits in 16 GB).
Escalation names the options, not their conditions.	If the escalation said “24 GB vs. 80 GB”, nobody would need to open a page and verification would be undefined.	“Two pages give different GPU options: kestrel-gpu (RTX 3090) or peregrine-gpu (A100). I’ll use kestrel-gpu.” Deciding requires opening a page and checking it against one’s situation; in provenance-only, opening a page is the only way to notice at all.
Say why the agent does not ask up front.	A reviewer will ask why the agent does not collect the user’s situation first.	One sentence: the agent cannot know which user facts matter until retrieval reveals the fork mid-task; hence mid-task escalation.

Sharpen the C5 tension.	MiPP’s C5 says stay silent or downgrade when uncertain about the <i>user’s context</i> , which is exactly this case.	Stated in Related Work: here silence means quietly applying a default that is wrong for this user; provenance-only is C5 taken literally, and the study tests it.
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3 Citations and integrity

What	Why	How
Rebuild every BibTeX entry from the publisher page.	Thirteen wrong first names, each keeping the correct initial, is the signature of a model expanding initials. The instructor treats it as an integrity issue.	<code>references.bib</code> was rebuilt against Crossref, IEEE Xplore, Springer, ACM camera-ready reference lines, ICLR proceedings and arXiv, and is unchanged in v3. You must still open each publisher page and check it yourself before submitting.
Cite published versions; drop non-references.	Chakraborti is HRI 2019; FictionalQA is ICLR 2026; MiPP and Möhle need LNCS 16744, editors and DOIs. Course slides, the internal RAG guide, Sharma et al. and Clegg et al. do not belong.	All fixed or removed. Mishra et al. stays arXiv-only because no published version exists as of 28 Sept 2026; the entry and the text say so.
Fill the fields the tools could not confirm.	Two ACM article numbers (Bansal CHI ’21, Ravi & Sindhgatta CHI EA ’25) and the ICLR booktitle wording could not be fetched.	Search <code>TODO authors</code> in both files (four comments in the <code>.tex</code> , five in the <code>.bib</code>); add <code>articleno</code> from the ACM DL; confirm the Schemmer spelling (Crossref “Kuehl”, ACM DL “Kühl”).

4 Claims that overstated their sources

What	Why	How
Mishra et al. “non-escalating baseline” and “use ECE”.	They evaluate nothing; one-shot retrieval is a taxonomy class, and ECE is a <i>proposed</i> criterion under an open challenge.	Cited only for the human-in-the-loop pattern (“request disambiguation”) and for proposed criteria; described as a survey available only as a preprint.
FictionalQA “removes” the pretraining confound.	It reduces it, for the model that ran the check, and says nothing about human prior knowledge.	Now “reduces”; screening plus item selection plus logs handle participants’ prior knowledge (someone who knows a 3090 has 24 GB).
Möhle: “fluency inflates trust independently of accuracy”.	Their convincingness main effect was non-significant; the finding is an interaction with <i>stated</i> accuracy; actual accuracy was never varied.	Cited for the interaction and as the reason to log behavior; Bansal et al. 2021 carries the “explanations raise acceptance regardless of correctness” claim.
Chakraborti “supplies the procedure” for calibration.	Their dataset had no incorrect plans.	Cited in one sentence as precedent that students given the ground truth can judge an agent’s plans, with the caveat.
“MiPP has never been tested”.	MiPP ran a within-subjects $n=48$ study of its three forms.	Precise gap: MiPP compared bundled policy families on a system that is never wrong, so verification had no consequence and forms were not compared at fixed decision points.

5 Study design

Note added 10 October 2026. The paper draft (`Assets/paper/main.tex`) supersedes the design below on one point: task set is no longer tied to escalation form, nor arranged in a 3×3 Latin square of set and form. Each block holds three tasks, one from each set, so set is crossed with form; the pool holds nine forks, three per set, each in two twin versions, and the assignment of forks to forms and twins to participants rotates with the participant number (`system/escalate/rotation.py`). The target is $N = 12$, two participants per block order, giving 108 measured tasks.

What	Why	How
Review-and-submit trial with a dropdown-only run card.	Reproducing a procedure is slow and hard to score; typed answers need adjudication; the earlier draft left the participant’s task undefined.	Task card → assistant streams 3–4 cited steps (about 3 s each) → run card of 3–4 <i>dropdowns</i> (3–4 options each, drawn from the corpus) pre-selected with the assistant’s choices → submit. 60–90 s per item; participants never type; scoring is automatic (scored field = key); nothing is adjudicated.
Score one field, hide which one; do not ask “is the assistant correct?”.	A card that asked only for the forked value, or a correctness question, would announce the fork in every condition and erase the manipulation; people act on output rather than grade it.	The scored field’s options include both diverging options plus 1–2 plausible alternatives; the other fields are correct, uncontested decoys. Control items cite two agreeing pages so two chips never signal a fork and give a baseline page-opening rate.
Define the forms by two properties, not by MiPP’s families.	In the earlier draft the quiet form still had a “sources differ” badge.	Announced? Blocking? Modal: yes/yes (stream pauses, overlay, card disabled until a choice). Inline: yes/no (same sentence on the fork step, card pre-filled, ignorable). Provenance-only: no/no (plain step with two citation chips). Same steps, default, pages, controls and undo in all three. Figure 1 of the proposal shows all three for item G1.

6 items per block; three hypotheses; trimmed measures.	21 items did not fit an hour; five hypotheses and a dozen measures at $N=12-18$ reads as fishing.	4 flagged (2 default-correct, 2 default-wrong) + 2 control per block; 2-item practice block; about 45 min. H1 verification, H2 appropriate reliance (form $\times$ default-correctness), H3 cost. Measures: verified, scored-field decision, time, raw TLX, one intrusion item, manipulation check, ranking with reason. Cut: need for cognition, other MiPP-Eval items, dwell thresholds.
Stay within subjects; handle carry-over explicitly.	Between subjects needs about three times the participants (4–6 per condition at class scale).	Each critical fact in one item only; task sets from disjoint sections; tell everyone up front the assistant can be wrong; all six orders; position covariate; first-block descriptive check.
Frozen JSON item records; written difficulty matching.	Participants must see identical content; the Latin square assumes exchangeable task sets.	Record = situation, question, steps with citations, fork step index, options, default, card fields with options and pre-fill, scored field, key. Match sets on steps, card fields, question–page overlap band, fork type, fork position; pilot; item random effect.
Power in one sentence.	The power table cost a quarter page and said one thing.	“A paired t -test at $N=12$ detects only large participant-level effects ($d_z \approx 0.9$), which is why trial-level models are primary; H2 power is simulated from the pilot.”

6 Corpus and contamination

What	Why	How
Corpus: CSU CS systems documentation (SNA); sections still open; requirements fixed.	Which sections to use is undecided; the constraints on any choice are not.	Frozen snapshot; key derivable from snapshot + situation line; snapshot and item records released together; 18 + 2 items from three disjoint sections (e.g., GPU/jobs, access/remote connection, storage/software); a system many students have not used.
Ground the illustrative items in the real pages.	The instructor asked for a real corpus; the item table must not invent facts.	Table 1 of the proposal uses facts from the crawled pages (GPU Jobs, Partitions, Connecting to Falcon, SSH guide, Home Directories, Temporary Disk Space, FAQ, Monitoring Jobs). The pairings are labelled illustrative until the snapshot is frozen; re-check each one against the snapshot.
No authored older revisions, no dated pairs.	The date rule is gone, so no page needs to be written or back-dated.	Forks come from existing pages whose scopes differ; nothing is added to the corpus. Report how many forks the check found and how many survived the blind-vs-informed filter.

7 Population and recruitment

What	Why	How
Name the population and the route.	“Domain experts” and “new hires” had no route.	CSU students with no prior use of the chosen system; screening item; class data-collection form; realistic pool is new graduate students and upper-division undergraduates.
IRB only if publishing, and early.	Recruitment cannot start before approval.	Talk to Ortega in the first two weeks of October; if going for IRB, file before the pilot.

8 The agentic claim

Direct answer: does this need agentic RAG, and is extending MiPP necessary?

Strictly, no, the question does not require an agentic system. It requires a system-initiated, mid-task escalation with provenance and a fallible default; any RAG assistant with a fork detector that fires while producing a multi-step procedure satisfies that, and the outcomes are human behaviours that do not depend on the pipeline behind the trigger.

What “agentic” buys is a real decision point that a chatbot cannot produce. A chatbot answers the question as asked. Only a loop that retrieves per step and inspects its own retrieval can discover mid-task that the answer depends on a fact the user did not state, and choose to hand the decision back. That is the course’s “if a chatbot can do it” test, passed without agency being the IV, and it is stronger in v3 than in v2: escalating a date conflict was something the agent should have resolved itself, whereas escalating a context-dependent fork is the one case where handing back is correct. The word stays honest only while four things hold: the task is multi-step; the escalation is agent-initiated at the affected step; the trigger is the pipeline’s own recorded decision (the fork check: retrieval-score margin plus an LLM judge asked whether the chunks give different answers depending on a user attribute the question does not state), not a hand-seeded flag; and the loop is described in one figure and one paragraph. If any fails, say “a RAG assistant with system-initiated escalation.”

Extending MiPP is the frame, not the paper. Use MiPP for three things only: the constraints held constant across forms (C1 brevity, C2 one-step undo, C3 provenance); the C5 tension that provenance-only tests; and the perceived-

intrusion item for comparability. The forms are not MiPP’s families and are not claimed to be. The theory for the DVs comes from the CHI/CSCW/IUI reliance literature (Bućinca, Bansal, Vasconcelos, Schemmer) and McFarlane for interruption. Never write “first test of MiPP”.

9 MiPP positioning

What	Why	How
Do not map the forms onto MiPP’s families.	MiPP’s families differ in trigger and in action authority (direct alert acts then offers undo; confirmatory prompt asks first). Our modal asks first and our inline acts first, so a one-to-one mapping would be wrong.	One sentence: the forms are defined by announced/blocking; we take MiPP’s constraints, not its taxonomy.
Verification anchored to the item, not the initiation.	MiPP’s measure is undefined when nothing is announced, which is the provenance-only case.	“Verified” = opened a diverging page between item onset and submit, in every form.
McFarlane as interpretation only.	MiPP never maps its families onto McFarlane’s methods.	One sentence, stated as the authors’ reading: modal $\approx$ immediate, inline $\approx$ negotiated, provenance-only does not interrupt.

10 Implementation

What	Why	How
One web app, one condition flag.	The fork step must be the only thing that differs between blocks.	React or plain JavaScript in a desktop browser. Left: task card and run card; center: assistant pane with citation chips; right: source panel (page with the passage highlighted). Replay engine reads the JSON record. About two weeks.
Drafting pipeline, run offline.	The agentic claim needs the escalation to be the loop’s own decision.	Python: chunk by heading, embed, retrieve top- k per sub-step, fork check (margin + LLM judge); only items where it fired become flagged items; report its hit rate. About one week.
State that behavior is pre-generated and replayed.	Participants never meet a live agent. That is the fixed-advice paradigm of the reliance studies and is legitimate, but only if disclosed, and only if the escalation decisions are the pipeline’s own.	New paragraph in the Approach: pre-generated and replayed; local open-weight model at temperature 0 with a fixed seed; authors edit wording, never a decision or default; correctness set by the situation line; participants told responses were prepared in advance; scope excludes multi-turn interaction. Citations marked [verify] in red until checked.
Log everything as events.	Verification is the primary DV and must be logged, not self-reported.	Page opens and duration, control uses, dropdown changes, submitted card and its correctness, timestamps; SQLite, participant code only.

11 Length, mock-ups and AI disclosure

What	Why	How
Cut from eight pages to about five, including the mock-ups.	The rubric says about two pages excluding references and figure pages; you chose to target about five pages total with the required mock-ups, and v2’s eight was too long.	Cut: the power table (one sentence instead), the design-at-a-glance table, the forms table (the figure now carries it), the long McFarlane and Ravi paragraphs, and repeated statements of the design. Related work is about one page and closes with the gap. Body is pages 1–5; references end on page 6.
Add the mock-ups the rubric requires.	System Vision must show a mockup; the three forms must be seen on the same item.	Figure 1: item G1 at the fork under all three forms, as TikZ wireframes (task card with the situation/asked split, dropdown run card, assistant pane with steps and chips; modal overlay / inline flag / plain citations). Figure 2: the drafting loop with the fork check. Table 1: five illustrative items including a default-correct and a control item.
Name each tool and what it did.	Instructor C2/C3: name tools; if a tool produced bib entries, say so and say you verified them.	Use of AI names Claude Code (Anthropic) running Claude Opus 5.5 with Claude Fable 5.1 subagents and lists: critique, MiPP reading, citation verification, drafting, TikZ mock-ups; states that the authors verify every reference and revise the text. Add any tool used on the original draft (there is a TODO).

What we keep from the original

- **The name and the interest.** “Escalate or Answer” and human escalation in agentic RAG, now as a question.
- **Within-subjects, counterbalanced design** with paired analysis, now three conditions in a 3×3 Latin square.
- **Cognitive load and accuracy as DVs.** NASA-TLX per block; the scored field and the keep/change decision at each

fork.

- **One interface shell, pre-generated and replayed responses**, now described as records drafted by the deployed loop and frozen.
- **MiPP-shaped escalation** (brief, reversible, provenance): the constraints held identical across forms.
- **The FictionalQA blind-vs-informed check**, as a stimulus filter on real items.
- **Difficulty-matched task sets**, with the method written down.
- **Forced choice plus thematic coding**, as an end-of-session ranking with a free-text reason.
- **The “given the retrievals” idea**, transformed: novices are given the cited pages in a source panel, which is what lets them verify.

12 Prioritized checklist for checkpoint 2

1. **Verify all 13 references on the publisher pages**; fill the two ACM article numbers and the ICLR booktitle; remove every TODO authors comment once done. (Integrity item; do first.)
2. **Read the new proposal end to end** and change anything that is not your view. Confirm in particular: the title; the situation/asked split; dropdown-only cards with automatic scoring; $N=18$ target / 12 minimum; 6 items per block; three hypotheses; the Team paragraph and roles.
3. **Decide which SNA sections form the three task sets**, freeze the snapshot, and check every illustrative pairing in Table 1 against it (R1 and S1 especially); relabel or replace any whose scope statement is not on the page.
4. **Talk to Ortega about IRB** in the first two weeks of October; decide publish-or-not; if publishing, file before the pilot.
5. **Build the drafting pipeline first**, run it over about 30 candidate questions, and record where the fork check fired; that record is what makes “agentic” honest.
6. **Run the blind-vs-informed filter** on every candidate item and record the survivor count.
7. **Assemble three task sets** with the written matching procedure; pilot with 3–4 people; rebalance.
8. **Build the web app**: condition flag, replay of the JSON records, dropdown run card, source panel, event log. Test that page opens are logged identically in all three forms.
9. **Quote McFarlane (2002) from the paper** for the negotiated-vs-immediate sentence, and confirm the Ravi & Sindhgatta participant count before adding it.
10. **Add any other AI tools** used on the original draft to the Use-of-AI section.